

SYDNEY BOYS HIGH SCHOOL



YEAR 11 YEARLY EXAMINATION 2001

MATHEMATICS

EXTENSION

Time allowed – 2 hours

Examiner: C Kourtesis

DIRECTIONS TO CANDIDATES

- All questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Use a new booklet for each question.
If required, additional booklets may be obtained from the Examination Supervisor upon request.

Question 1 (16 marks)

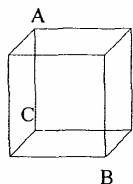
- (a) Express 120° in radians.
- (b) Find the value of $19^{-0.5}$ correct to two decimal places.
- (c) Factorize $8 - x^3$.
- (d) Find the gradient of the line $3x + 4y = 12$.
- (e) If $a^b = 3$, find the value of $a^{4b} - 5$.
- (f) Simplify $\frac{\log_2 32}{\log_2 16}$.
- (g) If $y = \frac{1}{x^2 + 3}$ find $\frac{dy}{dx}$.
- (h) If $f(\theta) = \theta^4 + 4\theta^{3/2}$ find the value of $f'(\theta)$.
- (i) Find the remainder when the polynomial $P(x) = x^3 - 4x$ is divided by $(x + 3)$
- (j) Given the equation of the parabola $(x - 2)^2 = 16(y + 4)$ find the:
- (i) coordinates of the vertex
 - (ii) coordinates of the focus
 - (iii) equation of the directrix

Question 2 (15 marks)

- (a) (i) Sketch the graphs of $y = x + 1$ and $y = |x - 2|$ on the same diagram.
- (ii) Find the coordinates of any points of intersection.

- (b) The graph of $y = 10(x + 1)(x - 3)$ intersects the x axis at two points A and B . Find the length of the line segment AB .

(c)



AB is a diagonal of a cube of side x cm and C is a vertex on the base of the cube. Determine the size of $\angle ABC$ correct to the nearest minute.

- (d) The point (x, y) is determined by the parametric equations $x = 3t - 1$ and $y = t^2 - 2t$ where $t \geq 0$. Find the value of x corresponding to $y = 3$.
- (e) Find the values of a , b , c and d if the following equation is satisfied by all real values of x .
- $$(ax + b)(x^2 + 3c) = x^3 - 2x^2 + 6x + d.$$
- (f) If $\sec \theta = \sqrt{5}$ and $\frac{\pi}{2} < \theta < 2\pi$, find the exact value of $\tan 2\theta$.

Question 3 (18 marks)

- (a) The interval AB has end points $A(-2,7)$ and $B(8,-8)$.

Find the coordinates of the point which divides AB internally in the ratio $2:3$.

- (b) If α, β and γ are the roots of the equation

$$x^3 - 2x^2 + 1 = 0$$

find the:

- (i) value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$
- (ii) equation whose roots are $-\alpha, -\beta, -\gamma$

- (c) Solve the inequality

$$\frac{2}{x-1} \leq 1$$

- (d) (i) Write down the expansion of $\sin(A+B)$.
- (ii) Solve for θ in the interval $0 \leq \theta \leq 2\pi$

$$\sin 2\theta = \frac{1}{2}$$

- (iii) Write down the general solution of

$$\sin 2\theta = \frac{1}{2}$$

- (e) The polynomial $P(x) = x^3 - 6x^2 + kx + 14$ has a zero at $x = 1$. Find the:

- (i) value of k .
- (ii) linear factors of $P(x)$.
- (iii) roots of the equation $P(x) = 0$.

Question 4 (15 marks)

- (a) If $Ax - 6y - 6B = 0$ and $7x - 18y + 9 = 0$ represent the same straight line, find the values of A , B and C .
- (b) Find the equation of the parabola if its axis is $y = 0$ and the line $y = x + 3$ is a tangent to the curve.
- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. M is the midpoint of the chord PQ . The tangents at P and Q meet at the point N .
- (i) Show that the equation of the tangent at P is $y - px + ap^2 = 0$ and write down the equation of the tangent at Q .
- (ii) Find the coordinates of M and N .
- (iii) Show that MN is parallel to the Y axis.
- (iv) Find the coordinates of T , the midpoint of MN .
- (v) Show that T lies on the parabola.

Question 5 (16 marks)

(a) If $F(x) = \log_2 \left(\frac{x+1}{x} \right)$ and $F(1) + F(2) + F(3) = \log_2 K$

(K a constant) find the value of K .

(b) The normal to the curve $y = \frac{ax+b}{\sqrt{x}}$

(a, b constants) has equation $4a + y = 22$ at the point where $x=4$. Find the values a and b .

(c) The equation $x^3 + 3Hx + L = 0$ has two equal roots. Prove that

$$L^2 + 4H^3 = 0$$

(d) A man travelling along a straight flat road passes three points at intervals of 200m. From these points he observes the angle of elevation of the top of a hill to the left of the road to be respectively 30° , 45° and again 45° .

(i) Draw a diagram of the above.

(ii) Find the height of the hill.



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2001

YEARLY EXAMINATION

YEAR 11

Mathematics Extension

Sample Solutions

$$(1) (a) 180^\circ = \pi^\circ$$

$$1^\circ = \frac{\pi}{180}$$

$$120 = \frac{\pi}{180} \times 120 = \frac{2\pi}{3} \quad (1)$$

$$(b) 0.23 \quad (1)$$

$$(c) 8 - x^3 = (2 - x)(4 + 2x + x^2) \quad (2)$$

$$3 + 4x + 2x^2 - 4x - 2x^2 - x^3$$

$$(d) 3x + 4y = 12$$

$$4y = -3x + 12$$

$$y = -\frac{3}{4}x + 3$$

$$m = -\frac{3}{4} \quad (1)$$

$$(e) a^{4b} - 5 = (a^b)^4 - 5$$

$$= 3^4 - 5 = 76 \quad (1)$$

$$(f) \frac{\log_2 2^5}{\log_2 2^4} = \frac{5 \log_2 2}{4 \log_2 2} = \frac{5}{4} \quad (2)$$

$$g) y = (x^2 + 3)^{-1}$$

$$y' = -1(x^2 + 3)^{-2} \times 2x = \frac{-2x}{(x^2 + 3)^2} \quad (2)$$

$$(h) f(\theta) = \theta^4 + 4\theta^{\frac{3}{2}}$$

$$f'(\theta) = 4\theta^3 + 4 \times \frac{3}{2} \theta^{\frac{1}{2}}$$

$$= 4\theta^3 + 6\theta^{\frac{1}{2}} \quad (1)$$

$$(i) \text{ let } P(x) = x^3 - 4x$$

$$\text{let } x + 3 = 0$$

$$x = -3$$

$$P(-3) = (-3)^3 - 4(-3)$$

$$= -27 + 12$$

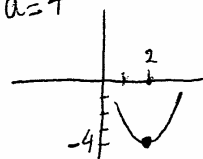
$$= -15 \quad (2)$$

$$(j) (x-2)^2 = 16(y+4)$$

$$(x-h)^2 = 4a(y+k)$$

$$(i) V(2, -4) \quad (1)$$

$$a = 4$$

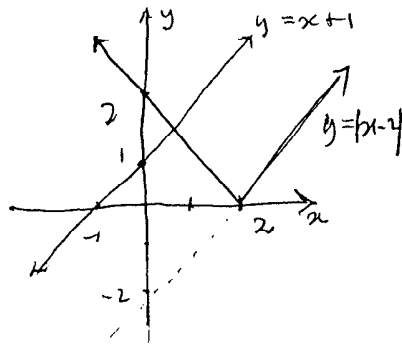


$$(ii) F(2, 0) \quad (1)$$

$$(iii) y = -8 \quad (1)$$

Question 2

a) (i)



$$(1) |x-2| = x+1$$

Square both sides

$$x^2 - 4x + 4 = x^2 + 2x + 1$$

$$3 = 6x$$

$$x = \frac{1}{2}$$

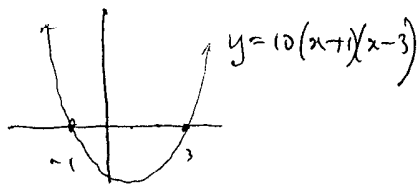
$$\text{Test: LHS} = \left| \frac{1}{2} - 2 \right|$$

$$= \frac{3}{2}$$

$$\text{RHS} = \frac{3}{2}$$

$\therefore \left(\frac{3}{2}, \frac{5}{2} \right)$ is intersection

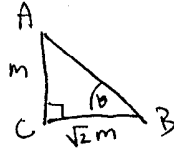
b)



Clearly $AB = 4$.

c) Base diagonal $BC^2 = m^2 + m^2$

$$BC = \sqrt{2}m$$



$$\tan \theta = \frac{m}{\sqrt{2}m}$$

$$= \frac{1}{\sqrt{2}}$$

$$\therefore \theta = 35^\circ 16'$$

(d) $x = 3t - 1$, $y = t^2 - 2t$ $t > 0$

$$\text{When } y = 3 \quad t^2 - 2t - 3 = 0$$

$$(t+1)(t-3) = 0$$

$$t = 3 \text{ or } -1$$

$$\therefore t = 3$$

$$\text{So } x = 3 \times 3 - 1$$

$$\underline{x = 8}$$

$$e) (ax+b)(x^2+3c)$$

$$= ax^3 + 3acx + bx^2 + 3bc$$

$$= ax^3 + bx^2 + 3acx + 3bc$$

$$\equiv x^3 - 2x^2 + 6x + d$$

Equating coefficients:

$$x^3: \underline{a = 1}$$

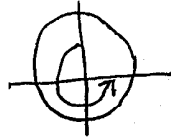
$$\underline{b = -2}$$

$$3ac = 6 \quad \therefore \underline{c = 2}$$

$$3bc = d \quad \therefore d = 3 \times (-2) \times 2$$

$$\underline{d = -12}$$

$$\textcircled{1} \sec \theta = \sqrt{5}$$

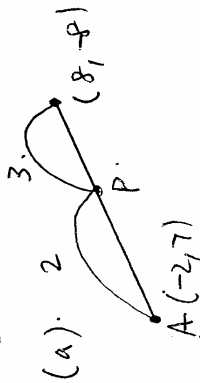


$$\cos \theta = \frac{1}{\sqrt{5}} \therefore \text{Ist Quad}$$

$$\theta \doteq 63.435^\circ$$

$$\therefore \tan \theta \doteq -\frac{4}{3}$$

3



$$x = \frac{2 \times 8 + 3(-2)}{2+3}$$

$$= 2$$

$$y = \frac{2(-8) + 3(7)}{5}$$

$$= 1 \quad (2, 1)$$

(b) $x^3 - 2x^2 + 1 = 0$

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= 0$$

(ii) $x^3 + (\sum \alpha_i)x^2 + (\sum \alpha_i \alpha_j)x + (-\alpha\beta\gamma) = 0$

Let $y = -x$

$$x = -y$$

$$(-y)^3 - 2(-y)^2 + 1 = 0$$

$$-y^3 - 2y^2 + 1 = 0$$

$$\therefore y^3 + 2y^2 - 1 = 0$$

is the equation

(c) $\frac{2}{x-1} \leq 1$

this inequality has boundaries given by

(i) $x = 1$

(ii) $2 = x - 1, x = 3$

~~$x = 0, 2, 4$~~

$$x = 0, -2 \leq 1$$

$$x = 2, 2 > 1$$

$$x = 4, 2/3 < 1$$

$$\therefore \{x : x < 1 \text{ or } x \geq 3\}$$

(d) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

(ii) $\sin 2\theta = \frac{1}{2}$

$$2\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

(iii) $\sin 2\theta = \sin \frac{\pi}{6}$

$$2\theta = n\pi + (-1)^n \frac{\pi}{6}$$

$$\therefore \theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$$

(e)

$$P(1) = 0$$

$$1 - 6 + k + 14 = 0$$

$$k = -9$$

(ii)

$$x-1 \left| \begin{array}{r} x^2 - 5x - 14 \\ x^3 - 6x^2 - 9x + 14 \\ \hline -(x^3 - x^2) \end{array} \right.$$

$$\frac{-5x^2 - 9x}{-(-5x^2 + 5x)} = \frac{-14x + 14}{-14x + 14}$$

$$\therefore P(x) = (x-1)(x-7)(x+2)$$

(iii) $P(x) = 0, x = 1, 7, -2$

4

(C) $x^2 = 4ay$

(1) $\frac{x^2}{4a} = y$
 $\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$ at $x=p$
 $y - ap^2 = \frac{p^2}{2a}(x - ap)^{m=p}$
 ~~$2ap \cdot \frac{p}{2a}$~~
 $y - ap^2 = px - ap^2$ (2)
 $y - px + ap^2 = 0$
 at x : $y - px + ap^2 = 0$

(II) $M: [a(p+q), \frac{a(p^2+q^2)}{2}]$ (1)

N: $y - px + ap^2 = y - qx + aq^2$
 $qx - px = ap^2 - aq^2$
 $x(q-p) = a(q+p)(q-p)$
 $x = a(q+p)$

So $y - ap(q+p) + ap^2 = 0$
 $y - apq - ap^2 + ap^2 = 0$ (2)
 $y = apq$

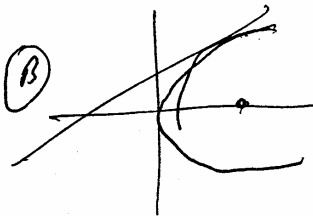
$N: [a(q+p), apq]$

(III) x coord of M & N $a(p+q)$ (1)
 $\therefore MN$ is parallel to x -axis

Contd T
 (IV) $MN: [a(p+q), \frac{a(p^2+q^2)}{2} + apq]$ (1)

(V) $[a(p+q), \frac{ap^2+aq^2+2apq}{4}]$ (3)

(A) $A = \frac{2}{3}$ (1)
 $B = -\frac{1}{2}$ (1)



$y^2 = 4ax$
 $y = x + b$
 Substitute
 $(x+b)^2 = 4ax$
 $x^2 + 2bx + b^2 = 4ax$
 $x^2 + (2b-4a)x + b^2 = 0$
 $\Delta = 0$ $(2b-4a)^2 - 4b^2 = 0$
 $16a^2 - 8ab = 0$
 $a = 2b$

$y^2 = 2x$ (2)

Contd T
 $[a(p+q), \frac{a(p^2+q^2)}{4}]$

$x = a(p+q)$
 $\frac{x^2}{4a} = p+q$

$y = \frac{a(p+q)^2}{4}$

$y = \frac{a(\frac{x}{a})^2}{4}$

$4y = a \cdot \frac{x^2}{a^2}$

$4y = \frac{x^2}{a}$

$x^2 = 4ay$

$\therefore$ Then on parabola

q5

(a) $F(x) = \log_2\left(\frac{x+1}{x}\right)$

$$\log m + \log n = \log(mn)$$

$$F(1) + F(2) + F(3) + F(4) = \log_2\left(\frac{2}{1}\right) + \log_2\left(\frac{3}{2}\right) + \log_2\left(\frac{4}{3}\right)$$

$$= \log_2\left(\frac{2}{1} \times \frac{3}{2} \times \frac{4}{3}\right)$$

$$= \log_2 4$$

$$\therefore k = 4$$

(b) $y = \frac{ax+b}{\sqrt{x}} = a\sqrt{x} + \frac{b}{\sqrt{x}} = ax^{\frac{1}{2}} + bx^{-\frac{1}{2}}$

$$\therefore y' = \frac{1}{2}ax^{-\frac{1}{2}} - \frac{b}{2}x^{-\frac{3}{2}}$$

$$x = 4, \quad y' = \frac{1}{2}a \times 4^{-\frac{1}{2}} - \frac{b}{2}(4)^{-\frac{3}{2}}$$

$$= \frac{a}{4} - \frac{b}{16}$$

$$= \frac{4a-b}{16}$$

$$y = \frac{4a+b}{2}$$

$$\therefore m_{\perp} = \frac{-16}{4a-b} = \frac{16}{b-4a} = \text{gradient of normal}$$

$$4x + y = 22 \quad \therefore m = -4$$

$$\boxed{x=4, y=6}$$

$$\therefore \frac{16}{b-4a} = -4$$

$$\& \quad \frac{4a+b}{2} = 6$$

$$\therefore \begin{array}{l} 16 = -4b + 16a \quad -(1) \Rightarrow 4 = -b + 4a \\ 12 = b + 4a \quad -(2) \Rightarrow 12 = b + 4a \end{array} \quad +$$

$$\therefore 8a = 16$$

$$a = 2 \Rightarrow b = 4$$

$$\boxed{a=2, b=4}$$

$$(c) \quad x^3 + 3Hx + L$$

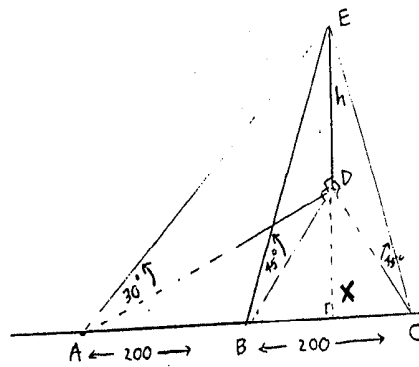
Let the roots be α, α, β

$$\left. \begin{aligned} \alpha + \alpha + \beta &= 0 \\ \alpha^2 + \alpha\beta + \alpha\beta &= 3H \\ \alpha^2\beta &= -L \end{aligned} \right\}$$

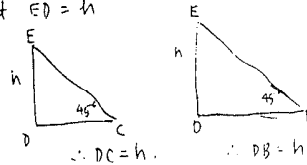
$$\left. \begin{aligned} 2\alpha + \beta &= 0 & \Rightarrow \beta &= -2\alpha \\ \alpha^2 + 2\alpha\beta &= 3H & \Rightarrow \alpha^2 + 2\alpha(-2\alpha) &= 3H \Rightarrow -3\alpha^2 = 3H \Rightarrow \alpha^2 = -H \\ \alpha^2\beta &= -L & \Rightarrow \alpha^2(-2\alpha) &= -L \Rightarrow -2\alpha^3 = -L \\ & & \Rightarrow \alpha^3 &= \frac{L}{2} \Rightarrow \alpha = \left(\frac{L}{2}\right)^{\frac{1}{3}} \end{aligned} \right\}$$

$$\left. \begin{aligned} \therefore \left[\left(\frac{L}{2}\right)^{\frac{1}{3}}\right]^2 &= -H \\ \therefore \left(\frac{L}{2}\right)^{\frac{2}{3}} &= -H \\ \therefore \left(\frac{L}{2}\right)^2 &= (-H)^3 = -H^3 \\ \therefore L^2 &= -4H^3 \\ \therefore L^2 + 4H^3 &= 0 \end{aligned} \right\} \quad \square \text{ED}$$

(d)

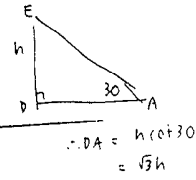


Let $ED = h$



$\therefore DC = h$

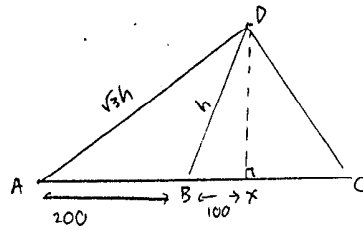
$\therefore DB = h$



$\therefore DA = h \cot 30^\circ = \sqrt{3}h$

$\therefore \triangle DBC$ is isosceles.

$\therefore BX = 100$



$$\therefore DX^2 + 100^2 = h^2 \quad \text{--- (1)}$$

$$DX^2 + 300^2 = 3h^2 \quad \text{--- (2)}$$

$$\therefore 2h^2 = 300^2 - 100^2$$

$$h^2 = \frac{300^2 - 100^2}{2}$$

$$\therefore h = \sqrt{\quad}$$

$$\therefore = 200\text{m}$$

(Pythagoras' Theorem)

$\therefore$ The height of the hill is 200 m